

4. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = bx_1^2 + a(x_2^2 + x_3^2) + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} b & 0 & 0 \\ 0 & a & b \\ 0 & b & a \end{pmatrix}$$

(2 marks) Its principal submatrices are:

$$\begin{aligned} 3^{\text{rd}} \text{ Order: } & \begin{pmatrix} b & 0 & 0 \\ 0 & a & b \\ 0 & b & a \end{pmatrix} \\ 2^{\text{nd}} \text{ Order: } & \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}, \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} \text{ and } \begin{pmatrix} a & b \\ b & a \end{pmatrix}, \\ 1^{\text{st}} \text{ Order: } & (b), (a) \text{ and } (a). \end{aligned}$$

- (b) (7 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *indefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is indefinite?

Solutions: (3 marks) $Q(\mathbf{x})$ is indefinite if $Q(\mathbf{x}') > 0$ for some $\mathbf{x}'$ and $Q(\mathbf{x}'') < 0$ for some $\mathbf{x}''$.

(4 marks) Thus, $Q(\mathbf{x})$ is indefinite if it is neither positive semidefinite nor negative semidefinite. $Q(\mathbf{x})$ is positive semidefinite if all principal minors are non-negative. $Q(\mathbf{x})$ is negative semidefinite if all odd principal minors are non-positive and all even principal minors are non-negative.

- (c) (14 marks) Find the values of the parameter values, a and b , such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? Then, plot these regions in $\mathbb{R}^2$.

Solutions: (9 Marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are non-negative which requires:

$$a \geq 0, \quad b \geq 0, \quad ab \geq 0, \quad a^2 - b^2 \geq 0, \quad a^2b - b^3 \geq 0.$$

For these to hold at once it must be that

$$a \geq 0, \quad b \geq 0, \quad a \geq b.$$

$Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

$$a \leq 0, \quad b \leq 0, \quad ab \geq 0, \quad a^2 - b^2 \geq 0, \quad a^2b - b^3 \leq 0.$$

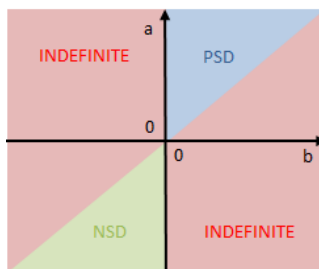
For these to hold at once it must be that

$$a \leq 0, \quad b \leq 0, \quad b \geq a.$$

Thus, the quadratic form is indefinite if and only if

$$ab < b^2.$$

The plot below depicts the answer



5. (25 marks) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a *quasi-concave* function and $g : \mathbb{R} \rightarrow \mathbb{R}$ a *weakly increasing* function,

$$x > y \Rightarrow g(x) \geq g(y).$$

Let $h(x) = g(f(x))$ for all $x \in \mathbb{R}$.

- (a) (7 marks) Provide two equivalent definitions of *quasi-concavity*.

Solutions: (1 mark) Let f be a function defined on a convex set $U \subseteq \mathbb{R}$. (3 marks) The function f is quasi-concave if upper-contour sets $C_k^+ = \{x | f(x) > k\}$ are convex for all values of k . (3 marks) The function f is quasi-concave if for all $x, y \in U$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \geq \min\{f(x), f(y)\}.$$

- (b) (10 marks) Is $h : \mathbb{R} \rightarrow \mathbb{R}$ *quasi-concave*? Prove your answer.

Solutions: Yes, h is indeed quasi-concave. Take any two points $x, y \in \mathbb{R}$ and any $t \in [0, 1]$. Quasi-concavity of f implies that

$$f(tx + (1 - t)y) \geq \min\{f(x), f(y)\}.$$

But if so, we must have that

$$\begin{aligned} h(tx + (1 - t)y) &= g(f(tx + (1 - t)y)) \geq g(\min\{f(x), f(y)\}) \\ &= \min\{g(f(x)), g(f(y))\} = \min\{h(x), h(y)\}, \end{aligned}$$

where the inequality and the following equality hold as $g' \geq 0$.

- (c) (8 marks) If x^* maximizes h , does it also maximize f ? If $\hat{x}$ maximizes f , does it also maximize h ?

Solutions: (2 marks) The set of maximizers of h contains the maximizers of f .

(3 marks) To show the latter, first observe that x^* maximizing h does not imply that x^* also maximizes f . For instance, if f has a unique maximum $\hat{x}$ and we consider a constant function $g(x) = 0$ (which is weakly increasing), any point in $\mathbb{R}$ other than $\hat{x}$ maximizes h , but not f .

(3 marks) In contrast, $\hat{x}$ maximizing f does indeed imply that $\hat{x}$ also maximizes h . To show the latter, note that $f(\hat{x}) \geq f(x)$ for $x \in \mathbb{R}$ indeed implies

$$h(\hat{x}) = g(f(\hat{x})) \geq g(f(x)) = h(x),$$

where the inequality holds as the transformation is increasing.

6. (50 marks) For $a \in (0, \infty)$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + y \\ \text{s.t.} \quad & x^2 + y^2 \leq 1 \\ & x \geq 0, y \geq ax. \end{aligned}$$

- (a) (4 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither.

Solutions: (2 marks) The objective function, as well as the last two constraints are obviously linear and thus both convex and concave. (2 marks) The first constraint is strictly convex, as its Hessian is positive definite for $x, y \geq 0$,

$$D^2g(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad |D^2F(x, y)| = 4 > 0 \text{ and } 2 > 0.$$

- (b) (4 marks) Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of $a > 0$ is the Slater condition satisfied?

Solutions: (4 marks) When the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case for all $a < \infty$ as the point $\check{x} = \varepsilon > 0$ and $\check{y} = (1 + a)\varepsilon$ is in the interior of the constraint set for $\varepsilon < (1/(1 + (1 + a)^2))^{1/2}$, as

$$\check{x}^2 + \check{y}^2 = \varepsilon^2(1 + (1 + a)^2) < 1, \quad \check{x} > 0, \quad \check{y} > a\check{x}.$$

- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangean?

Solutions: (6 marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= a - 2x\lambda_1 + \lambda_2 - a\lambda_3, \\ 0 &= 1 - 2y\lambda_1 + \lambda_3, \\ 0 &= \lambda_1(x^2 + y^2 - 1), \\ 0 &= \lambda_2x = \lambda_3(y - ax), \\ 0 &\geq x^2 + y^2 - 1, \quad 0 \geq -x, \quad 0 \geq ax - y, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(2 marks) Any solutions to these conditions is maximizer of the Lagrangean as the Lagrangean is a sum of concave functions by the arguments in (a) and (b).

- (d) (16 marks) Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (2 marks) At an optimum it also cannot be that $\lambda_1 = 0$, as the FOC for good y would imply that, $\lambda_3 = -1 < 0$. So there is no solution on these supports.
- (2 marks) At an optimum it also cannot be that $\lambda_2, \lambda_3 > 0$ as it would imply $x = y = 0$ and $\lambda_1 = 0$ which cannot be by the previous argument.
- (2 marks) At an optimum it also cannot be that $\lambda_1, \lambda_2 > 0$ while $\lambda_3 = 0$. If so, $x = 0$ and the FOC for good x would imply that

$$\lambda_2 = -a < 0.$$

- (6 marks) Next consider the case in which $\lambda_1, \lambda_3 > 0$ while $\lambda_2 = 0$. If so, the two binding constraints would imply that

$$y(a) = ax(a) \quad \text{and} \quad x(a) = \frac{1}{(1+a^2)^{1/2}}.$$

If so, as $x(a) > 0$, solving the two FOC for goods x and y would imply that

$$\lambda_1(a) = \frac{2a}{(1+a^2)^{1/2}} \geq 0, \quad \lambda_3(a) = \frac{a^2-1}{1+a^2}.$$

But then $\lambda_3(a) \geq 0$ would require $a \geq 1$. Thus, there is a solution on this support if and only if $a \geq 1$.

- (6 marks) In the final scenario, the optimum must live in the only remaining support, namely $\lambda_1 > 0$ and $\lambda_3 = \lambda_2 = 0$. Solving FOC an the binding constraint at such values one finds that

$$y(a) = \frac{1}{(1+a^2)^{1/2}}, \quad x(a) = ay(a), \quad \lambda_1(a) = \frac{1}{2y(a)} > 0.$$

For this to be a solution it must be that

$$y(a) \geq ax(a) \quad \Leftrightarrow \quad y(a) \geq a^2y(a) \quad \Leftrightarrow \quad 1 \geq a.$$

Thus, there is a solution on this support if and only if $a \leq 1$.

- (e) (10 marks) Exploit the Envelope Theorem to find the slope of value function with respect to a . State the result and apply it to this setting.

Solutions: The Envelope Theorem in general states that if $(\mathbf{x}(a), \boldsymbol{\lambda}(a))$ are differentiable functions and that $x(a)$ satisfies the constraint qualification. If so,

$$\frac{dv(a)}{da} = \frac{\partial L(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = x(a)(1 - \lambda_3(a))$$

If $a \leq 1$ and thus $\lambda_3(a) = 0$, we have that the value increases in a

$$\frac{dv(a)}{da} = x(a) = \frac{a}{(1+a^2)^{1/2}} > 0.$$

If $a \geq 1$ and thus $\lambda_3(a) > 0$, the slope of the value with respect to a amounts to

$$\frac{dv(a)}{da} = x(a)(1 - \lambda_3(a)) = \frac{2}{(1+a^2)^{3/2}} > 0.$$

- (f) (6 marks) For what value of $a \in [0, 10]$ is the value function maximized?

Solutions: To maximize the value with respect to a , we solve

$$\max_{a \in [0, 10]} v(a).$$

As $dv(a)/da > 0$ for all $a \in (0, 10]$, the solution must set $a = 10$.